Vermax

NSE swing-trading decision platform

Technical documentation: system design, data, prediction engine, API and UI

Version 0.2 | 1 October 2026

Stack: NestJS (TypeScript) API | Python 3.14 + LightGBM ML engine | React 19 + Vite UI | SQLite
Scope: daily end-of-day picks with stop-loss and targets, sector priority (Defence & Engineering), fundamentals and news overlays, screeners (stocks and IPOs), monitoring and alerts, journal, backtest.

Decision-support software. Not investment advice. Backtested edge is small; losses are normal.

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Reading guide. Sections 2-4 explain how the system is built and how a pick is produced. Sections 7-8 are the interface contract. Section 9 states what the backtest does and does not prove.

1. Product overview

Vermax is a daily decision-support application for Indian equity swing trading (holding period up to 10 trading days). After the NSE publishes end-of-day data, it ranks about 1,000 liquid stocks with a machine-learning model, applies quality and risk filters, and publishes a short list. Every pick carries a ready trade plan: entry rule, stop-loss, two targets, position size and the reasons behind the ranking.

Primary user needs

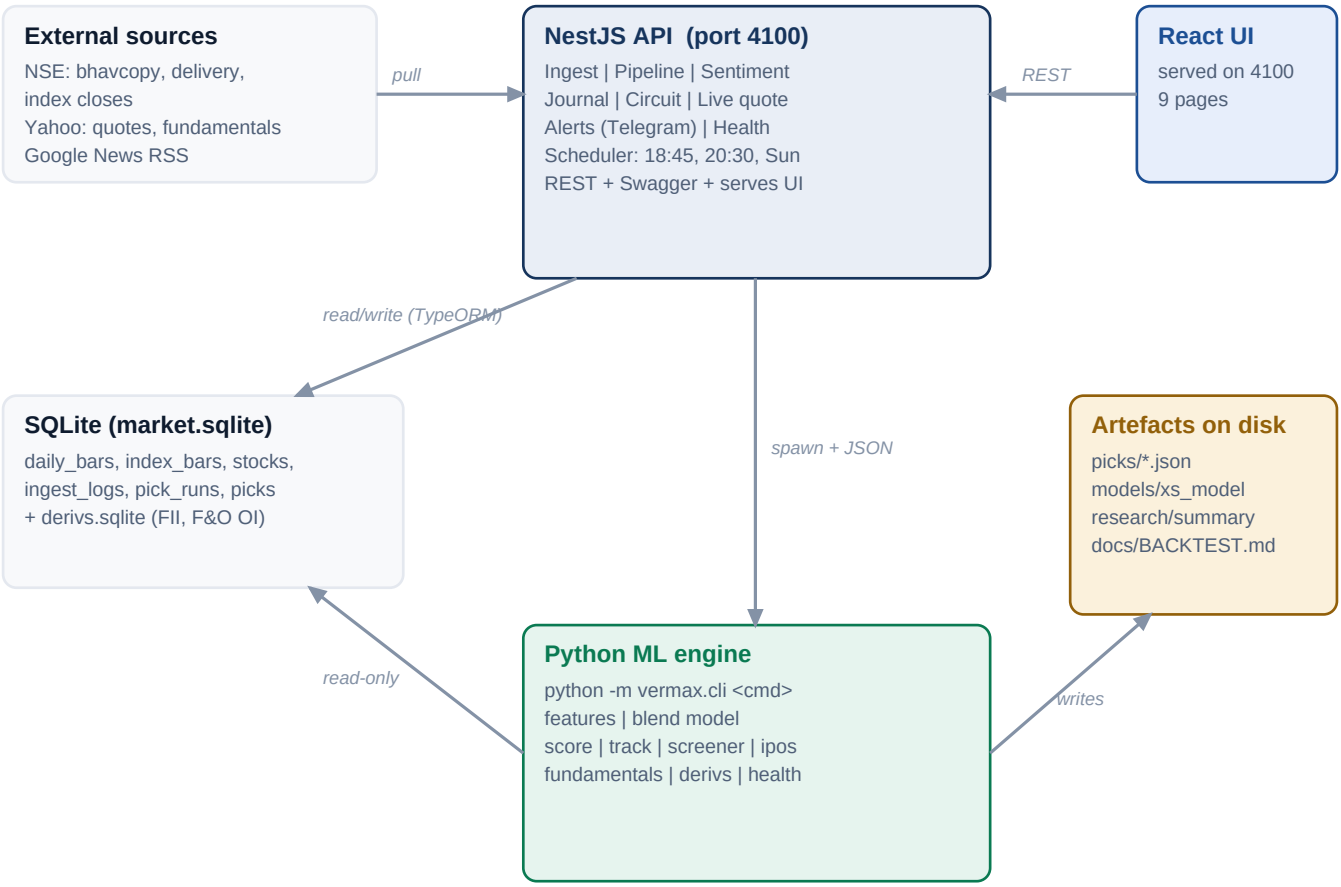
- A next-day list of stocks priced Rs200-650, with stop-loss and 3%-15% target distances, holding at most 10 sessions.
- Strong emphasis on **Defence and Engineering** sectors, applied only while the sector itself is strong.
- Fundamentals (ROE, ROCE, debt, growth), news sentiment, candle and market-regime awareness in one place.
- A stricter weekly **Elite 3** list, a price-range screener, lower-circuit recovery history and a live paper-trade journal.

Design principles

- **Evidence over optimism.** Every rule that ships is backtested walk-forward; overlays that cannot be backtested are labelled as such.
- **Official numbers.** All analytics use NSE end-of-day closes; live quotes are display-only and labelled delayed.
- **One trade-plan definition.** The backtest, the published plan and the live tracker share the same state machine.
- **Separation of concerns.** NestJS orchestrates and serves; Python owns every numeric and ML computation.

The realistic goal is a small positive expectancy per trade with controlled risk. A steady 5-12% every week is not achievable by this or any system, and the product says so on its Backtest and Elite pages.

2. System architecture



Component	Technology	Responsibility
API (apps/api)	NestJS 12, TypeORM, better-sqlite3, node-cron, fast-xml-parser	Ingest, orchestrate the pipeline, sentiment, journal, circuit analysis, live quote, REST + Swagger, scheduling
ML engine (ml/vermax)	Python 3.14, pandas 3, numpy, LightGBM, yfinance, pyarrow	Features, model training, scoring, trade simulation, fundamentals, screener, backtest report
UI (apps/web)	React 19, Vite 8, Tailwind 4, TanStack Query, Recharts, react-router	Decision screens; all data via the REST API; light enterprise theme
Storage	SQLite (WAL) + JSON/parquet files	Market data and published runs in SQLite; model, research and cache artefacts on disk

Integration pattern: JSON CLI bridge

NestJS never links against Python. It spawns `python -m vermax.cli <command> '<json>'` and reads one JSON document from the last stdout line. Commands: `train, score, track, fundamentals, screener`. The database path is passed by environment variable and Python opens it read-only, so the API remains the only writer of market tables. This keeps the two runtimes independent and each command individually testable from a terminal.

Ports and processes

Process	Default	Notes
Vermax (API + UI)	4100	UI at /, API at /api, Swagger at /docs; run by scripts/start-vermax.cmd with auto-restart
Web dev server	5190	Development only; Vite proxies /api
Python	on demand	Spawned per command; 20 s to 10 min depending on command

3. Data layer

3.1 Sources

Source	Data	Use	Timing
NSE archives - bhavcopy (UDiFF and legacy)	Daily OHLC, previous close, volume, traded value, trades (EQ series)	Prices and liquidity for all analytics	~18:30 IST
NSE archives - MTO	Delivery quantity and delivery %	Delivery-surge feature	Can lag bhavcopy
NSE archives - index close	Nifty 50, India VIX, sector and thematic indices	Market regime features	Can lag bhavcopy
NSE - Nifty 500 list, equity master	Company names, industry	Sector mapping	Weekly cache
Yahoo Finance	Fundamentals, statements, earnings date; delayed intraday quote	Fundamentals overlay; display-only live price	On demand, cached
Google News RSS	Headlines (10-day window)	Sentiment overlay	Cached 3 h

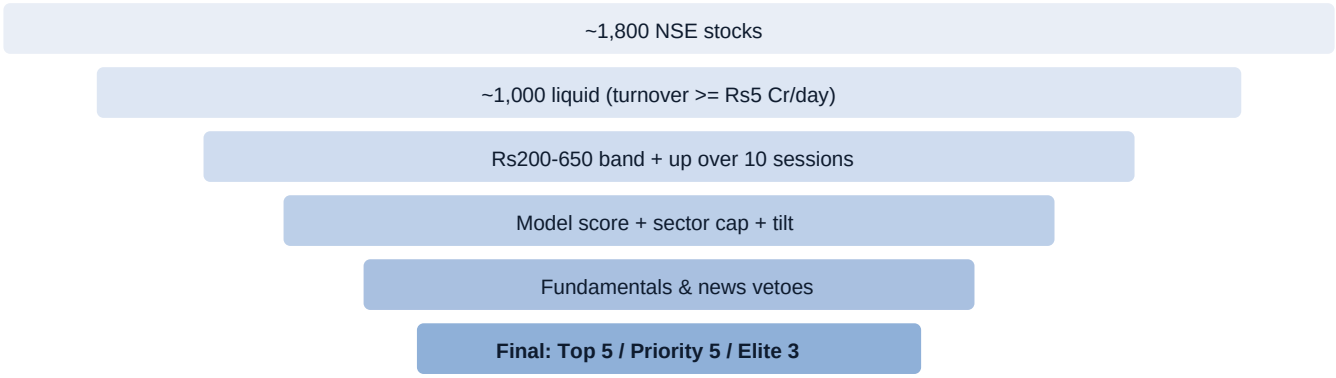
3.2 Storage schema (market.sqlite)

Table	Key columns	Purpose
daily_bars	symbol, date, open, high, low, close, prevClose, volume, turnover, trades, deliveryPct	3.4 M+ rows since 2019; unique (symbol, date)
index_bars	indexName, date, close, changePct, pe, pb	Nifty, VIX and ~190 indices
stocks	symbol, name, isin, sector, industry, inNifty500, first/lastDate	Symbol master and sector map
ingest_logs	date, source, status (OK/EMPTY/ERROR), rows, message	Ingestion audit and retry decisions
pick_runs	asOf (unique), regime, payload (JSON), createdAt	One published run per signal date, including rankings
picks	asOf, list (core/theme/elite), symbol, plan levels, status, entry, netRet, exitReason	Forward paper-trade journal

3.3 Ingestion rules

- **Idempotent per date:** a session is replaced inside a transaction; raw files are cached on disk so backfills never re-download.
- **Holiday vs not-yet-published:** an EMPTY result is only final once the IST calendar day has passed the trading date.
- **Partial-day repair:** if index or delivery files were not yet published when the bhavcopy was loaded, the next run re-fetches those recent sessions automatically.
- **Corporate actions:** price history is back-adjusted for splits and bonuses by detecting previous-close ratios that match canonical factors (1/2, 1/5, 1/10, 2, 5 ...).
- **Point-in-time universe:** liquidity (20-day average traded value >= Rs5 Cr), price and history filters are evaluated per row, not on today's constituents.

4. Prediction engine



4.1 Features (about 70 per stock-day)

Group	Examples
Trend and momentum	1/5/10/20/60/120-day returns, 12-1 month momentum, distance to 20/50-EMA and 200-DMA, EMA slopes, RSI(14), ADX and +DI/-DI, strength vs Nifty
Volatility and structure	ATR%, Bollinger width and squeeze ratio, range contraction, 30-day base depth, distance to 20-day and 52-week high
Participation	Volume vs 20-day average, delivery % and delivery surge
Candle shape	Close position in range, body, wicks, gap, hammer, bullish engulfing, NR7, inside bar
Event setups	20-day breakout, pullback to 20-EMA, squeeze breakout, 52-week-high breakout, number of setups firing
Market regime	Nifty returns and distance to 50/200-DMA, drawdown, India VIX level/percentile, breadth (% above 50/200-DMA), advance ratio
Sector and theme	Equal-weight sector basket returns, sector strength rank, sector breadth, stock strength within sector, Defence / Engineering flags

4.2 Model

- **Algorithm:** blend of a 5-seed LightGBM regressor ensemble and a LightGBM LambdaRank ranker; the score is the average of their daily percentile ranks (0-100). Each model: 250 rounds, 31 leaves, depth 6, L2 = 50, feature and bagging fraction 0.7.
- **Target:** cross-sectional percentile rank of the 10-day forward return (next-open entry to close of day 10) among all liquid stocks that day. Ranking relative strength makes the model market-neutral by construction.
- **Why not deep learning:** daily end-of-day data (about 7 years) is too small; LSTM/Transformer price models overfit and fail live. Tabular boosting is the sound baseline.
- **Explainability:** TreeSHAP contributions (pred_contrib) produce the per-pick 'why' reasons shown in the UI.

4.3 Validation

- **Anchored walk-forward:** the model for year Y is trained only on data before 1 January of Y minus a 15-day embargo; predictions for Y are out-of-sample.
- **Metrics:** rank information coefficient by year (0.048 to 0.088, positive every year 2021-2026), score-decile spread, expectancy, profit factor, drawdown, weekly return distribution.
- **Baseline:** random picks with identical exits are always reported alongside the model.

4.4 Selection rules

Step	Rule
Universe	Liquid (>= Rs5 Cr/day), price Rs200-650, not circuit-locked (|1-day move| < 19%)
Momentum gate	Return over the last 10 sessions must be positive
Ranking	Model score plus 0.03 for Defence/Engineering stocks whose sector ranks >= 60th percentile among all sectors
Diversification	Maximum 2 picks per sector
Overlays	Reject on weak fundamentals (< 30/100), loss-making, D/E > 2, ROE < 5%, results within 7 days, or a severe negative headline
Elite 3	Above 50-EMA and 200-DMA, up over 10 and 20 days, sector >= 60th percentile, fundamentals PASS, news not negative; at most 3, fewer if fewer qualify

4.5 Trade plan (single shared state machine)

Element	Definition
Entry	Buy at the next session open; skip if it opens more than 2.5% above the signal close
Stop-loss	1.5 x ATR, clamped to 3%-6% below entry. Checked before targets each bar; gap-through fills at the open
Target 1	+1.5R: book 50%, move stop to breakeven from the next bar
Target 2	+3R: exit the remainder (typically +9% to +18%)
Time stop	Exit at the close of trading day 10
Sizing	Risk 0.5% of capital / stop distance, capped at 20% of capital per stock; halved when Nifty is below its 200-DMA
Costs (backtest)	0.40% round trip: brokerage, STT, exchange/SEBI/stamp/GST, slippage

5. Overlays and analytics

These modules refine or explain the model output. Fundamentals and news use free sources without point-in-time history, so they are **live risk filters only** and are not part of the backtest.

5.1 Fundamentals quality score

Points-based, transparent scoring out of 100 (financial-sector stocks skip ROCE, debt and cash-flow tests):

Metric	Test	Points
ROE (3-year average)	>= 15%	25
ROCE	>= 15%	15
Debt / equity	<= 0.5	15
Sales CAGR	>= 10%	15
Profit CAGR	>= 10%	15
Operating cash flow	positive	10
Promoter / insider holding	>= 35%	5

Verdict: PASS (>= 60), WATCH, AVOID (score < 30, loss-making, D/E > 2, ROE < 5%, results within 7 days) or UNKNOWN when fewer than three metrics are available.

5.2 News sentiment

Google News RSS headlines from the last 10 days are scored by a finance-specific regex lexicon (order wins, profit changes, upgrades, resignations, promoter selling). Scores are recency-weighted (half-life about 3 days) and normalised to -1..+1. Governance or legal red flags (fraud, SEBI action, ED/CBI raid, insolvency, auditor exit) veto a pick.

5.3 Lower-circuit history

The NSE end-of-day file has no circuit flag, so a lower-circuit day is inferred: close equals the day's low and the fall from the previous close matches a 5%, 10% or 20% band. Consecutive locked days merge into one event. For each event the system reports sessions to regain half and all of the pre-circuit price, the further fall over the next 10 sessions, and aggregate rates (recovered within 10 and 30 sessions).

5.4 Sector strength and priority

Equal-weight baskets of liquid stocks form sector indices; strength rank is the percentile of the 20-day return among sectors. Defence and Engineering are maintained as explicit themes. Backtest evidence shows Defence alone is not reliably strong, so the priority bonus switches on only while the theme's own rank is >= 60th percentile.

5.5 Screener

Scans about 1,700 NSE equities (ETFs excluded via the NSE equity master) for positive returns over any combination of 1W, 2W, 3W, 4W (5/10/15/20 sessions) and 1M, 3M, 6M, 1Y (calendar months, first close on or after the same date, matching Google). Default: positive in every period and price at most Rs600. A Live prices toggle recomputes the visible rows from the delayed intraday price using live-anchored reference prices, so intraday returns match Google; filters always use the official NSE close. Price ranges (Rs50-100, 100-150, 150-200, 200+) are multi-select. Each row shows the model buy rank, an A-D grade (rank percentile) and a Tomorrow rating tag: Elite, Top pick, Priority pick, Eligible or Not eligible with reasons.

5.6 IPO screener

Recent main-board listings use the official listing date from the NSE equity master (EQUITY_L). For each: listing price (first traded price), return since listing, returns over every period available since listing (n/a otherwise), distance from the high since listing, turnover, and the same live-price mode. Filters: listed within 1M to 3Y, max price, minimum turnover, positive in every available period, above listing price. The IPO issue price is not in NSE public archives, so it is not shown.

5.7 Derivatives positioning (context)

NSE participant-wise open interest (FII, DII, Client, Pro) and stock F&O; open interest are stored since 2019 in derivs.sqlite. A walk-forward A/B test showed a small IC gain but a worse worst-year and 2026, so these are shown as market context (for example the FII index-futures long ratio) and are not model inputs.

6. Daily pipeline

#	Step	Owner	Output
1	Ingest missing sessions; repair sessions lacking index or delivery data	NestJS DataService	daily_bars, index_bars, ingest_logs
2	Update open paper trades with the latest bars	NestJS Journal -> Python track	picks status, entry, exit, P&L;
2b	Refresh FII and F&O; open interest (last 20 days)	Python derivs	derivs.sqlite
3	Build features and score the liquid universe; build Top, Priority and Elite candidate pools	Python score	data/picks/<date>.json
4	Fundamentals for the pooled symbols (5-day cache)	Python fundamentals	quality score, verdict, flags
5	News sentiment for the same symbols (3-hour cache)	NestJS Sentiment	score, label, severe flag
6	Apply vetoes, choose final lists, compute position sizes, attach lower-circuit summary	NestJS PicksService	PublishedRun
7	Persist the run and one paper trade per published symbol	NestJS	pick_runs, picks
8	Refresh screener and IPO caches; send the picks summary alert	Python + NotifyService	screener/*.json, Telegram

Scheduling and triggers

- Weekdays 18:45 IST: full pipeline. 20:30 IST: retry if data or picks are not current. 21:00 IST: watchdog alert if the session is still missing. Sunday 09:00 IST: retrain, then a model-health check.
- Manual: `POST /api/pipeline/run` or the Refresh button in the UI; progress is exposed at `GET /api/pipeline/state`.
- Re-running the same signal date replaces earlier untouched picks that are no longer published.

Monitoring and alerts

- **Model health** (GET /api/model/health, Backtest page): live rank-IC of each published ranking versus the realised 10-session return, monthly out-of-sample IC, and input drift (PSI for stock-level features; historical percentile for market-level features). Status OK, WATCH or ALERT.
- **Alerts** (Telegram, optional via TELEGRAM_BOT_TOKEN and TELEGRAM_CHAT_ID): daily picks summary, pipeline failures with the failing step, missing-data warnings, retrain results and health warnings. Recent events also appear in the sidebar.
- **System status** (GET /api/system): NSE data date, published picks date, next scheduled jobs, alert configuration.

Failure handling: the pipeline is single-flight; a failed step records the error message in state. Python output is sanitised so a partially published data day (NaN values) cannot corrupt the JSON contract.

7. API reference

Base path `/api`. JSON responses. Interactive documentation at `/docs` (Swagger).

Method and path	Purpose	Key response fields
GET /health	Liveness	ok, time
GET /picks/latest	Latest published run (rankings omitted)	asOf, tradeDate, regime, core[], theme[], elite[], rejected[], sectors[], universe
GET /picks/:asOf	Run for a signal date	same as latest
GET /picks/history	Recent runs	asOf, regime, createdAt

Method and path	Purpose	Key response fields
POST /pipeline/run	Start the pipeline (single-flight)	started, state
GET /pipeline/state	Progress	running, step, error, lastAsOf
GET /sectors	Sector strength table	sector, rank, returns, breadth, top names
GET /screener	Multi-horizon scan with rating	rows[] with d5..y1, modelRank, rating{grade, tomorrow, eligible, reasons}
GET /stock/:symbol[?deep=1]	Stock detail: 420 bars, model rank, circuit history; deep adds fundamentals and news	bars[], model, circuit, fundamentals, sentiment
GET /quote/:symbol	Delayed live quote (display only)	price, changePct, dayHigh, dayLow, time
GET /journal	Paper-trade log and statistics	stats{winRate, expectancy, PF}, picks[] with circuit summary
POST /journal/refresh	Recompute open trades	updated, stats
GET /data/status	Data coverage and ingest log	firstDate, lastDate, tradingDays, recentLogs
POST /data/ingest	Manual ingest (optional from/to)	storedDays, holidays, errors
GET /performance	Backtest summary	variants[], ic, deciles, elite{}
GET /ipos	Recent listings with period returns	rows[] with listingDate, listOpen, sinceListing, d5..y1, refs, refsLive
GET /quotes?symbols=	Batch delayed live quotes (max 60)	symbol -> quote
GET /model/health	Model health and drift	status, reasons, liveIC[], oosMonthlyIC[], drift[]
GET /system	Operational status	dataLatest, publishedAsOf, jobs[], alerts
POST /system/test-alert	Send a test Telegram alert	delivered, telegram

Pick object (core / theme / elite)

```
symbol, sector, score, rank, close, changePct, plan{entryRef, maxEntry, stop, stopPct, t1, t1Pct,
t2, t2Pct, maxHoldDays}, reasons[], tags[], decision (BUY/WATCH), fundamentals{verdict,
qualityScore, roe, roce, debtToEquity, ...}, sentiment{score, label, items[]}, circuit{events,
meanRecoveryDays, ...}, sizing{qty, capitalUsed, riskAmount}.
```

8. User interface

Single-page React application with a left sidebar (Tomorrow, Elite 3, Screener, IPOs, Sectors, Journal, Backtest, symbol search, system status and the Refresh button). Light enterprise theme, Inter typeface, tabular numerals, full-width content.

Page	Contents	Behaviour
Tomorrow	Trade date, market regime strip, capital and risk calculator, Top picks, Defence and Engineering list, vetoed stocks	Cards show buy limit, stop, T1, T2, size and risk; details expand on demand; info icon explains the recommendation on hover
Elite 3	Rules checklist, up to three Elite picks, realistic-expectation statistics from the backtest	Shows fewer than three, or none, when checks fail
Screener	1W-1Y period toggles and presets, max price (Rs600), min turnover, search, multi-select price ranges, live prices	Tomorrow rating, buy rank, veto flag; ascending/descending sort; pagination
IPOs	Recent listings, listing price, since-listing return, every period since listing, from-high	Listed-within filter, positive in every period, above listing price, live prices
Sectors	Priority theme cards, all sectors ranked by strength	Explains whether the ranking bonus is currently active
Journal	Win rate, expectancy, profit factor; tracked picks with status and average lower-circuit recovery	Update outcomes on demand
Backtest	Strategy comparison, yearly returns, rank-IC, decile chart, plan definition	Includes the honest reading of results
Stock	NSE close, delayed live price, chart with 5D-1Y ranges, return tiles, EMA/DMA/volume toggles, lower-circuit history, fundamentals and news	Calendar-month returns; labels state which close the numbers use

Front-end design notes

- Server state through TanStack Query with a 60 s stale time; capital and risk settings persisted in localStorage with safe fallbacks.
- Position size is recomputed client-side from the published stop distance, so changing capital never needs a pipeline run.
- Charts use Recharts with animations disabled for instant range changes; all colours come from CSS design tokens.

9. Validation results

Out-of-sample window 1 January 2021 to 2 September 2026, walk-forward, 0.40% round-trip costs, published trade plan, 0.5% risk per trade, at most 5 open positions, production blend model.

Strategy	Trades	Expectancy / trade	Profit factor	CAGR	Max drawdown
Random 5 liquid stocks/day, same exits	6,895	-0.25%	0.91	-1.3%	-27.2%
Model top-5, no tilt	6,903	+0.34%	1.13	7.8%	-17.5%
Model top-5 + tilt, no live filters	6,903	+0.43%	1.17	10.5%	-12.1%
Live rules: Rs200-650, 2-week positive, tilt (published)	6,901	**+0.24%**	**1.09**	**13.1%**	**-14.6%**

Key findings

- Ranking skill is real but modest: rank-IC 0.051 (2021), 0.057, 0.068, 0.090, 0.068, 0.069 (2026).
- Model experiments (same walk-forward): single regressor IC 0.063, 5-seed ensemble 0.064, LambdaRank 0.062, blend 0.067 with the best worst-year IC and lowest drawdown, so the blend is in production. Run-to-run noise in expectancy is about 0.1% per trade.
- Elite 3 (weekly, 857 trades): 35% of picks reach Target 1, 45% are stopped, +0.55% per trade, at least one of three reaches T1 in 65% of weeks, all three lose in 20% of weeks. It matches plain model top-3; its value is quality gating.
- With the expanded Engineering theme (AI and data-centre infra, EPC, power and cables) and the blend model: Defence +0.67% and Engineering +0.94% per model pick. Name lists use present-day knowledge, so this is optimistic.
- Adding confirmation filters (volume, RSI band, delivery, candle position) did not raise the hit rate, which stayed near 30-36%.
- Requiring 2-week positive momentum lowered expectancy from +0.43% to +0.24% (before the price band) but matches the user's risk preference. Halving risk to 0.5% cut simulated drawdown from about -28% to -15%.

Interpretation. Roughly one in three picks reaches Target 1 and about half hit the stop. The edge is a slightly favourable payoff earned across many trades, not a guarantee on any stock. The tilt weight was chosen after viewing out-of-sample results and theme name lists use today's knowledge, so those gains are optimistic.

10. Operations and configuration

Run

```
npm install
python ml/seed_from_legacy.py <legacy.sqlite> data/market.sqlite (one-off)
cd ml && python -m vermax.cli train (about 10 min)
npm run build -w web && npm run build -w api
scripts\start-vermax.cmd (UI + API on :4100, auto-restart)
scripts\install-autostart.cmd (start at Windows logon)
```

Setting (env)	Default	Meaning
PORT	4100	API port
VERMAX_DB	data/market.sqlite	SQLite path (also passed to Python)
PICKS_MIN_PRICE / PICKS_MAX_PRICE	200 / 650	Price band for all published lists
PICKS_CORE / PICKS_THEME	5 / 5	List sizes
CAPITAL / RISK_PER_TRADE	500000 / 0.005	Default sizing (UI overrides per user)
TELEGRAM_BOT_TOKEN / TELEGRAM_CHAT_ID	unset	Enable alerts (scripts/.env.cmd)
RETRY_CRON / WATCHDOG_CRON	30 20 * * 1-5 / 0 21 * * 1-5	Retry and missing-data watchdog
DAILY_CRON / RETRAIN_CRON / TZ_NAME	45 18 * * 1-5 / 0 9 * * 0 / Asia/Kolkata	Scheduler
SCHEDULER_ENABLED	true	Disable cron jobs
PYTHON_BIN	python	Interpreter used by the bridge

Files and testing

- data/picks/<date>.json (scored run), data/models/xs_model.txt (+ meta), data/research/* (features, OOS predictions, summary.json), data/fundamentals/*.json, data/screener/*.json.

- docs/BACKTEST.md is regenerated by `python -m vermax.report` after `python -m vermax.xsection`.
- Unit tests (Vitest, 17 passing): sentiment lexicon, position sizing, session dates, lower-circuit detection and recovery.
- The Python trade simulator was verified for exact parity between the refactored backtest and the live tracker.

11. Limitations and roadmap

Known limitations

- End-of-day data only: no intraday signals; live quote is delayed and display-only.
- Fundamentals, news and lower-circuit inference are not backtested (no point-in-time history; circuit flag inferred).
- Survivorship: Defence/Engineering name lists reflect today's knowledge; index membership is not point-in-time.
- Regime dependence: 2025-26 were flat to negative for the model. Recent regime shifts can persist.
- Price band Rs200-650 excludes most large Defence names (HAL, BEL, BDL, Mazagon), so the Defence list is often empty.

Roadmap (ordered by expected value)

Priority	Item	Rationale
Done	Regressor ensemble + LambdaRank blend; monitoring, drift and alerts; auto-restart and autostart; 0.5% default risk	Delivered in v0.2
Done (context)	FII/DII participant OI and stock F&O; OI since 2019	Tested; not adopted as model inputs
1	Results calendar and bulk/block deals (needs a reliable historical source)	Event-risk and smart-money signals
2	FinBERT / LLM news scoring	Better live sentiment than the lexicon
3	Point-in-time index membership	Removes survivorship bias from research
4	Broker integration (GTT orders)	Only after 50+ closed journal trades confirm the edge

Acceptance rule for every change: it must improve the walk-forward out-of-sample numbers against the random baseline and the current published strategy, and then survive at least 50 closed trades in the live journal.

Regulatory note: publishing these picks to other people requires SEBI Research Analyst registration in India. This document describes software for personal decision support.